

PAUL-CHRISTIAN BÜRKNER

GENERAL INFORMATION

<i>Date of Birth</i>	16 June 1991
<i>Place of Birth</i>	Marburg, Germany
<i>Work Address</i>	Vogelpothsweg 87, 44227 Dortmund, Germany
<i>Email</i>	paul.buerkner@gmail.com
<i>Website</i>	https://paulbuerkner.com

KEY SCIENTIFIC METRICS

<i>Publications</i>	144 (peer-reviewed only)
<i>Funding</i>	2,253,000 € (third-party only)
<i>Citations</i>	45,385 (source: GoogleScholar)
<i>h-index</i>	52 (source: GoogleScholar)

WORK EXPERIENCE

TU Dortmund University	<i>since 2023</i>	Full Professor of Computational Statistics Full Professor of Computational Statistics, Department of Statistics, TU Dortmund University, Germany.
	<i>2020-2023</i>	Independent Junior Research Group Leader Independent Junior Research Group Leader for Bayesian Statistics at the Cluster of Excellence SimTech, University of Stuttgart, Germany.
University of Stuttgart	<i>2019-2020</i>	Postdoctoral Researcher Postdoctoral researcher at the chair of Computational Probabilistic Modeling (Prof. Dr. Vehtari), Aalto University, Department of Computer Science, Finland.
	<i>2014-2019</i>	Research Associate Research associate at the chair of Statistics and Methods (Prof. Dr. Holling), University of Münster, Department of Psychology, Germany.

HIGHER EDUCATION

PhD in Psychology	<i>2014-2017</i>	University of Münster Grade: Summa Cum Laude · Institute of Psychology Title: <i>Optimal Design and Bayesian Data Analysis</i> . Received multiple awards (see the Awards section).
	<i>2014-2017</i>	University of Hagen Grade: 1.3 · Institute of Mathematics Thesis: <i>On the Statistics of Curie–Weiss–Distributed Random Variables</i> .
	<i>2013-2014</i>	University of Münster Grade: 1.1 · Institute of Psychology Thesis: <i>Adaptive Designs for Logistic Models with False Answers</i> .
Master of Mathematics	<i>2011-2014</i>	University of Hagen Grade: 1.7 · Institute of Mathematics Thesis: <i>A Hull Operator for Complex Matroids</i> .
	<i>2010-2013</i>	University of Münster Grade: 1.2 · Institute of Psychology Thesis: <i>Testing for Publication Bias in Diagnostic Meta-Analysis: A Simulation Study</i> .

RESEARCH FUNDING

GACR & DFG	Modrak M. & Bürkner P. C. (2026). <i>grammo: A grammar of probabilistic models</i> . <i>Funder: Czech Research Foundation (GACR) and German Research Foundation (DFG)</i> . Total: 556,000 €. Own share: 338,000 € .
MERCUR	Bürkner P. C., Hoffmann D., & Kobeleva X. (2026). <i>Deep Bayes for Biomedicine</i> . <i>Funder: MERCUR Foundation</i> . 300,000 € .
DFG	Fried et al. (2024). <i>Spatio-temporal Statistics for the Transition of Energy and Transport</i> (Collaborative Research Center). <i>Funder: German Research Foundation (DFG)</i> . Total: 12,540,000 €. Own share: 312,000 € .
DFG	Bürkner P. C. & Radev S. T. (2023). <i>BayesFlow: Simulation Intelligence with Deep Learning</i> . <i>Funder: German Research Foundation (DFG)</i> . 353,000 € .
DFG	Bürkner P. C. (2022). <i>Intuitive Joint Priors for Bayesian Multilevel Models</i> . <i>Funder: German Research Foundation (DFG)</i> . 238,000 € .
DFG	Bürkner P. C. (2022). <i>Bayesian Distributional Latent Variable Models</i> . <i>Funder: German Research Foundation (DFG)</i> . 238,000 € .
DFG	Bürkner P. C. & Bulling A. (2022). <i>Amortized Bayesian Inference for Multilevel Models</i> . <i>Funder: German Research Foundation (DFG)</i> . 232,000 € .
EXC SimTech	Guthke A. & Bürkner P. C. (2022). <i>Data-Integrated Training of Surrogate Models for Uncertainty Quantification and Diagnostics of Complex Biological Systems</i> . <i>Funder: Cluster of Excellence SimTech</i> . 285,000 € .
Cyber Valley	Bürkner P. C. (2021). <i>Meta-Uncertainty in Bayesian Model Comparison</i> . <i>Funder: Cyber Valley Research Fund</i> . 242,000 € .
EXC SimTech	Bürkner P. C. & Sedlmair M. (2021). <i>Machine Learning for Bayesian Model Building</i> . <i>Funder: Cluster of Excellence SimTech</i> . 285,000 € .
EXC SimTech	Bulling A. & Bürkner P. C. (2021). <i>Bayesian Intent Prediction for Human-Machine Collaboration</i> . <i>Funder: Cluster of Excellence SimTech</i> . 175,000 € .
ELLIS	Bulling A. Bürkner P. C., Kuchenbecker J. K., Pradel M., Schulte im Walde S., Staab S., Steinwart I., & Vu T. (2021). <i>Stuttgart ELLIS Unit</i> . <i>Funders: ELLIS Society and University of Stuttgart</i> .

SELECTED AWARDS

GCPR	2023 · Best paper honorable mention award at the German Conference of Pattern Recognition (GCPR).
SIPS	2020 · Mission award of the Society for Improving Psychological Science (SIPS) for brms.
SIPS	2020 · Commendation award of the Society for Improving Psychological Science (SIPS) for brms.
University of Münster	2018 · Award for the best dissertation 2017-2018 in Psychology at the University of Münster.
German Society for Psychology	2017 · Gustav A. Lienert Award for the best methodological dissertation in Psychology awarded by the German Society for Psychology (DGPs).
University of Münster	2017 · Award for the best lecture at the Institute of Psychology in Münster.
German National Acad. Foundation	2014 · Scholarship of the German National Academic Foundation (Studienstiftung des deutschen Volkes).

OPEN-SOURCE SOFTWARE

brms	Lead author · An R package for Bayesian regression models using Stan. Received multiple awards (see the Awards section).
posterior	Lead author · An R package for working with posterior distributions.
thurstonianIRT	Lead author · An R Package for fitting Thurstonian IRT models.

<i>BayesFlow</i>	Author · A Python library for simulation-based Bayesian inference.
<i>posterioradb</i>	Author · A Posterior Database for Bayesian Inference.
<i>elicitio</i>	Author · A Python package for expert prior elicitation.
<i>loo</i>	Author · An R package for approximate leave-one-out cross-validation.
<i>ggsimplex</i>	Author · An R package for simplex visualizations with ggplot2.
<i>bayesim</i>	Author · An R package for simulations with Bayesian models.
<i>bayehear</i>	Author · An R package for metrics to evaluate Bayesian models.
<i>bayesfam</i>	Author · An R package for custom brms families.
<i>bayesian</i>	Author · An R package to interface brms and tidymodels.
<i>rstan</i>	Contributor · An R Interface to Stan.
<i>bayesplot</i>	Contributor · An R package for visualizing Bayesian models.
<i>propred</i>	Contributor · An R package for projection predictive variable selection.
<i>emmeans</i>	Contributor · An R package for estimating marginal means.

SELECTED PROFESSORSHIP CALLS

<i>Full Professorship</i>	2022 · Call for the Full Professorship (W3) in Computational Statistics, Department of Statistics, TU Dortmund University, Germany. Accepted.
<i>Full Professorship</i>	2022 · Call for the Full Professorship (W3) in Data Analytics and Computational Statistics, Department of Computer Science, University of Konstanz, Germany.

SELECTED SCIENTIFIC INVOLVEMENT

<i>Area Chair</i>	since 2026 · Area chair of the Annual Conference on Neural Information Processing Systems (NeurIPS; https://neurips.cc).
<i>Associate Member</i>	since 2026 · Associate member of the Horst Görtz Institute for IT Security (https://hgi.rub.de/).
<i>Associate Member</i>	since 2026 · Associate member of the ELLIS Unit NRW.
<i>Editor</i>	since 2025 · Editor of the Journal of Robustness Reports.
<i>Member</i>	since 2024 · Member of the committee for the re-accreditation of Data Science studies at TU Dortmund University.
<i>Founding Member</i>	2024 · Founding member of the Computational Clinical Psychology and Psychotherapy Network (https://ccpp.network/) funded by the DFG.
<i>Chairman</i>	2024 · Chairman of the appointment committee for the associate professorship in Causality at TU Dortmund University.
<i>Reviewer</i>	since 2024 · Reviewer in appointment procedures for professorships.
<i>Organizer</i>	2022 · Organizer of the 1st International SimTech Summer School, University of Stuttgart. Co-Organizers: Benjamin Unger and Kristyna Pluhackova.
<i>Reviewer</i>	since 2022 · Reviewer for major funding agencies. <i>Selection:</i> German Research Foundation (DFG), European Research Council (ERC), Dutch Research Council (NWO), Humboldt Foundation.
<i>Member</i>	since 2021 · Member of the ELLIS Society (https://ellis.eu/).
<i>Member</i>	2021 – 2024 · Member of Cyber-Valley (https://cyber-valley.de/en).
<i>Faculty Member</i>	2021 – 2023 · Faculty Member of the International Max Planck Research School for Intelligent Systems (IMPRS-IS; https://imprs.is.mpg.de/).
<i>Founding Member</i>	2021 – 2023 · Founding member of the Stuttgart ELLIS Unit (https://ellis.eu/units/stuttgart).
<i>Member</i>	since 2018 · Member of the Stan Development Team (https://mc-stan.org/).
<i>Consultant</i>	since 2018 · Academic consultant in industry. Selection: Bayer (2018), Novartis (since 2021), Axem (since 2022).

Editor

2018 – 2020 · Associate editor of Biostatistics.

Reviewer

since 2014 · Reviewer for international journals and conferences. *Selection:* AISTATS, Bayesian Analysis, Behavior Research Methods, Biometrical Journal, ICML, ICLR, Journal of Machine Learning Research, Journal of Probability and Statistics, Journal of Statistical Software, Journal of the Royal Statistical Society, Nature, Nature Ecology & Evolution, Nature Human Behaviour, NeurIPS, Philosophical Transactions, Psychological Methods, Psychometrika, Psychonomic Bulletin and Review, Statistics in Medicine.

SELECTED TALKS

Multilevel
Conference

2026 · Utrecht · Keynote
Title: *Amortized Bayesian Inference for Multilevel Models.*

QUAMU
Conference

2026 · Bielefeld · Invited Talk
Title: *Bayes in the Age of Machine Learning.*

ELLIS
Unconference

2025 · Copenhagen · Invited Talk
Title: *Robust Amortized Bayesian Inference with Self-Consistency Losses.*

Bayescomp

2025 · Singapore · Contributed Talk
Title: *Robust Amortized Bayesian Inference with Self-Consistency Losses.*

One World ABI
Seminar

2025 · online · Invited Talk
Title: *Amortized Mixture and Multilevel Models.*

Cyber Valley

2025 · Stuttgart · Invited Talk
Title: *Meta-Uncertainty in Bayesian Model Comparison.*

University of
Bielefeld

2025 · online · Invited Talk
Title: *The future of Bayes and brms with application in biology.*

Stan Conference

2024 · Oxford · Contributed Talk
Title: *Generative Bayesian Modeling with Implicit Priors.*

PHYSTAT-SBI
Workshop

2024 · Munich · Invited Talk
Title: *A Statistical Perspective on Simulation-Based Inference.*

Bayes on the Beach
Conference

2024 · Gold Coast · Keynote
Title: *Does Bayes have to be slow? A glimpse into amortized Bayesian inference.*

Oxford University

2023 · Oxford · Keynote
Title: *Probabilistic Modeling for Ecology.*

Princeton
University

2023 · online · Invited Talk
Title: *An Introduction to Bayesian Statistics.*

DagStat
Conference

2022 · Hamburg · Contributed Talk
Title: *The sparse polynomial chaos expansion: a fully Bayesian approach with joint priors on the coefficients and global selection of terms.*

Psychoco
Conference

2021 · online · Keynote
Title: *Bayesian Item Response Models.*

Oslo UseR Group

2021 · online · Invited Talk
Title: *Bayesian multilevel modeling with brms.*

Oxford University

2020 · online · Invited Talk
Title: *Bayesian regression modeling.*

Turku University

2020 · online · Invited Talk
Title: *Bayesian multilevel modeling with brms.*

TU Dortmund
University

2020 · online · Invited Talk
Title: *Bayesian multilevel modeling with brms.*

Bayer

2020 · online · Invited Talk
Title: *Bayesian multilevel modeling with brms.*

Stat. Methods for
Linguistics

2019 · Potsdam · Keynote
Title: *A Principled Bayesian Workflow for Data Analysis.*

University of
Duisburg-Essen

2019 · Essen · Invited Talk
Title: *A Principled Bayesian Workflow for Data Analysis.*

DGPs Conference	2019 · Kiel · Contributed Talk Title: <i>Improving Convergence Diagnostics for MCMC Sampling Algorithms.</i>
Stan Conference	2019 · Cambridge · Contributed Talk Title: <i>Leave-Future-Out Cross-Validation for Bayesian Time-Series Models.</i>
Multilevel Conference	2019 · Utrecht · Keynote Title: <i>Bayesian Multilevel Modeling with brms and Stan.</i>
DagStat	2019 · Munich · Contributed Talk Title: <i>Leave-Future-Out Cross-Validation for Bayesian Time-Series Models.</i>
Stan Conference	2018 · Helsinki · Contributed Talk Title: <i>Custom Response Distributions in brms.</i>
EAM Conference	2018 · Jena · Contributed Talk Title: <i>Handling Ordinal Predictors in Regression Models via Monotonic Effects.</i>
Bayes@Lund	2018 · Lund · Keynote Title: <i>Why Not to be Afraid of Priors.</i>
DGPs Conference	2017 · Tübingen · Keynote Title: <i>Optimal Design and Bayesian Data Analysis.</i>
eRum Conference	2016 · Poznan · Contributed Talk Title: <i>brms: An R Package for Bayesian Multilevel Models using Stan.</i>
Int. Workshop on Simulation	2015 · Vienna · Contributed Talk Title: <i>Adaptive Designs for Logistic Models with False Answers.</i>
DGPs Conference	2015 · Jena · Contributed Talk Title: <i>Optimal Design of Non-Parametric Two-Sample Tests.</i>

SELECTED WORKSHOPS

Multilevel Conference	2026 · Utrecht · 1 day Title: <i>Bayesian modeling with brms.</i>
Oxford University	2023 · Department of Biology · 1 day Title: <i>Bayesian modeling for biologists using brms.</i>
University of Tübingen	2023 · Center of Methods · 2 days Title: <i>Bayesian modeling with the brms package.</i>
TU Dortmund University	2022 · Department of Statistics · 2 days Title: <i>Bayesian Statistics.</i>
University of Salzburg	2022 · Department of Psychology · 2 days Title: <i>Introduction to Bayesian Data Analysis.</i>
Oxford University	2021 · Department of Zoology · 4 days Title: <i>Bayesian Regression Modelling for Biologists.</i>
Research Cluster SMiP	2020 · Mannheim · 2 days Title: <i>Introduction to Stan: A Probabilistic Programming Language for Bayesian Inference.</i>
University of Aarhus	2020 · Department of Economics and Business Economics · 1 day Title: <i>Bayesian Model and Variable Selection.</i>
MPI for Human Development	2019 · Göttingen · 1 day Title: <i>Bayesian Multilevel Modeling.</i>
MPI for Emp. Aesthetics	2019 · Frankfurt · 2 days Title: <i>Bayesian Multilevel Modeling.</i>
Multilevel Conference	2019 · Utrecht · 1 day Title: <i>Introduction to Bayesian Data Analysis.</i>
DagStat Conference	2019 · Munich · 1 day Title: <i>Bayesian Data Analysis using Stan.</i>
University of Lausanne	2018 · Department of Psychology · 2 days Title: <i>Introduction to Meta-Analysis.</i>

University of Magdeburg	2018 · Department of Psychology · 4 days Title: <i>Introducing Basic and Advanced Bayesian Modelling</i> .
University of Aarhus	2018 · 4 days Title: <i>Advanced Bayesian Statistical Modeling</i> .
ETH Zurich	2018 · 1 day Title: <i>Classical and Bayesian Multi-Level Models in R</i> .
University of Hamburg	2017 · Department of Psychology · 2 days Title: <i>Fitting Multi-Level Models in R</i> .
DPPD Conference	2017 · Munich · 1 day Title: <i>Bayesian Multi-Level Models in R with brms</i> .
University of Bern	2017 · Department of Psychology · 3 days Title: <i>Bayesian Multi-Level Models in R with brms</i> .
University of Münster	2017 · Department of Psychology · 3 days Title: <i>Introduction to Bayesian Inference</i> .
University Paris Decardes	2017 · 1 day Title: <i>Introduction to Meta-Analysis</i> .
DGPs Conference	2016 · Leipzig · 1 day Title: <i>Bayesian Multilevel Models in R using the Package brms</i> .

SELECTED TEACHING ACTIVITIES

Data Science and Statistics	2025–2026 · TU Dortmund University · 2 times Lecture: <i>Simulation-Based Inference</i> .
Data Science and Statistics	2024–2026 · TU Dortmund University · 2 times Lecture: <i>Statistical Learning for Big Data</i> .
Data Science and Statistics	2024–2026 · TU Dortmund University · 3 times Lecture: <i>Case Studies I and II</i> .
Data Science and Statistics	2024–2026 · TU Dortmund University · 3 times Seminar: <i>Model Comparison</i> .
Data Science and Statistics	2023–2026 · TU Dortmund University · 3 times Lecture: <i>Applied Bayesian Data Analysis</i> .
Data Science and Statistics	2023–2026 · TU Dortmund University · 4 times Seminar: <i>Multilevel Models</i> .
Data Science and Statistics	2023 · TU Dortmund University · 1 times Lecture: <i>Computational Statistics</i> .
Simulation Science	2022 · University of Stuttgart · 1 times Lecture: <i>Bayesian Statistics and Probabilistic Machine Learning</i> .
Simulation Science	2021 · University of Stuttgart · 1 times Lecture: <i>ML Sessions: Bayesian Statistics</i> .
Simulation Science	2021–2022 · University of Stuttgart · 2 times Seminar: <i>Advanced Topics in Simulation Science</i> .
Psychology	2018 · University of Münster · 2 times Seminar: <i>Structural Equation Modeling and Bayesian Statistics</i> . Average Evaluation: 10.9 points (15 point <i>abitur</i> scale).
Psychology	2014–2019 · University of Münster · 5 times Lecture: <i>Descriptive Statistics and Probability Theory</i> . Average Evaluation: 12.6 points (15 point <i>abitur</i> scale). Award for the best lecture in the winter semester 2016/2017.
Psychology	2015–2018 · University of Münster · 4 times Lecture: <i>Inferential Statistics</i> . Average Evaluation: 12.1 points (15 point <i>abitur</i> scale).

CURRENT PHD STUDENTS

TU Dortmund University	since 2026 · Yiming Zang · Statistics Topic: <i>Deep Bayes for Biomedicine</i> .
TU Dortmund University	since 2026 · Hans Olischläger · Statistics Topic: <i>Improving Neural Architectures for Amortized Bayesian Inference</i> .
TU Dortmund University	since 2024 · Svenja Jedhoff · Statistics Topic: <i>Real-time Spatio-Temporal Data Analysis for Monitoring Logistics Networks</i> . Co-Advisor: Prof. Anne Meyer
TU Dortmund University	since 2024 · Aayush Mishra · Statistics Topic: <i>Robust and Efficient Learning in Amortized Bayesian Inference</i> .
TU Dortmund University	since 2024 · Lars Kühmichel · Statistics Topic: <i>BayesFlow: Simulation Intelligence with Deep Learning</i> . Co-Advisor: Prof. Stefan Radev
TU Dortmund University	since 2022 · Luna Fazio · Statistics Topic: <i>Bayesian Distributional Latent Variable Models</i> .

GRADUATED PHD STUDENTS

TU Dortmund University	2022 – 2026 · Florence Bockting · Statistics Topic: <i>Simulation-Based Prior Distributions for Bayesian Models</i> .
University of Tübingen	2022 – 2026 · Soham Mukherjee · Statistics Topic: <i>Probabilistic Models for scRNA Sequencing Data</i> . Co-Advisor: Prof. Manfred Claassen
TU Dortmund University	2021 – 2025 · Javier Aguilar · Statistics Topic: <i>Intuitive Joint Priors for Bayesian Multilevel Models</i> .
University of Stuttgart	2022 – 2025 · Philipp Reiser · Computer Science Topic: <i>Data-Integrated Training of Surrogate Models for Uncertainty Quantification and Diagnostics of Complex Biological Systems Models</i> . Co-Advisor: Dr. Anneli Guthke
University of Stuttgart	2021 – 2025 · Marvin Schmitt · Computer Science Topic: <i>Meta-Uncertainty in Bayesian Model Comparison</i> .
Aalto University	2019 – 2023 · Alejandro Catalonia · Computer Science Topic: <i>Robust Bayesian Methods for Model and Variable Selection</i> . Primary Advisor: Prof. Aki Vehtari
University of Münster	2018 – 2021 · Niklas Schulte · Psychology Topic: <i>Statistical Properties of Forced-Choice Questionnaires in Applicant Personality Measurements</i> . Primary Advisor: Prof. Heinz Holling

CURRENT POSTDOCTORAL RESEARCHERS

TU Dortmund University	since 2024 · Šimon Kucharský Topic: <i>Applications of Amortized Bayesian Inference</i> .
TU Dortmund University	since 2023 · Daniel Habermann Topic: <i>Amortized Bayesian Inference for Multilevel Models</i> .

FORMER POSTDOCTORAL RESEARCHERS

University of Stuttgart	2022 – 2024 · Lei Shi Topic: <i>Bayesian Intent Prediction for Human-Machine Collaboration</i> . Co-Advisor: Prof. Andreas Bulling
University of Heidelberg	2021 – 2023 · Stefan Radev Topic: <i>Amortized Bayesian Inference</i> . Became an assistant professor at Rensselaer Polytechnic Institute, Troy, USA.

ALL PUBLICATIONS

In Review

- 172) Aguilar J. E. & **Bürkner P. C.** (in review). Dependency-Aware Shrinkage Priors for High Dimensional Regression. *ArXiv preprint*.
- 171) Aguilar J. E., Kohns D., Vehtari A., & **Bürkner P. C.** (in review). R2 priors for Grouped Variance Decomposition in High-dimensional Regression. *ArXiv preprint*.
- 170) Bignardi G., Kievit R. A., & **Bürkner P. C.** (in review). A general method for estimating reliability using Bayesian Measurement Uncertainty. *PsyArXiv preprint*.
- 169) Bockting F. & **Bürkner P. C.** (in review). elicito: A Python Package for Expert Prior Elicitation. *ArXiv preprint*.
- 168) **Bürkner, P. C.**, Gabry, J., Kay, M., & Vehtari, A. (in review). posterior: Tools for Working with Posterior Distributions in R. *Local preprint*.
- 167) Fazio, L., & **Bürkner, P. C.** (in review). Latent Variable Models for Distributional Features. *ArXiv preprint*.
- 166) Fazio L., Scholz M., & **Bürkner P. C.** (in review). Primed Priors for Simulation-Based Validation of Bayesian Models. *ArXiv preprint*.
- 165) Jedhoff S., Kutabi H., Meyer A. & **Bürkner P. C.** (in review). Efficient Uncertainty Propagation in Bayesian Two-Step Procedures. *ArXiv preprint*.
- 164) Kucharský Š., Mishra A., Habermann D., Radev S. T., & **Bürkner P. C.** (in review). Improving the Accuracy of Amortized Model Comparison with Self-Consistency. *ArXiv preprint*.
- 163) Kühmichel, L., Huang, J. M., Pratz, V., Arruda, J., Olischläger, H., Habermann, D., Kucharský, Š., Elsemüller, L., Mishra, A., Bracher, N., Jedhoff, S., Schmitt, M., **Bürkner, P. C.**, & Radev, S. T. (in review). BayesFlow 2.0: Multi-Backend Amortized Bayesian Inference in Python. *ArXiv preprint*.
- 162) Mishra, A., Kucharský, Š., & **Bürkner, P. C.** (in review). Unsupervised Continual Learning for Amortized Bayesian Inference. *ArXiv preprint*.
- 161) Modrák M., Stroppel S., & **Bürkner P. C.** (in review). Simulation-based validation of Bayes factor computation. *ArXiv preprint*.
- 160) Mukherjee S., Aguilar J. E., Zago M., Claassen M., & **Bürkner P. C.** (in review). Latent variable estimation with composite Hilbert space Gaussian processes. *ArXiv preprint*.
- 159) Nalborczyk, L. & **Bürkner, P. C.** (in review). Precise temporal localisation of M/EEG effects with Bayesian generalised additive multilevel models. *biorXiv preprint*.
- 158) Revathe, T., Weidling, M. T., Utami-Atmoko, S. S., Setia, T. M., Razik, I., van Schaik, C. P., Whiten, A., **Bürkner, P. C.**, & Schuppli, C. (in review). Eight years of social and asocial learning synergistically shape orangutan diet profiles. *biorXiv preprint*.
- 157) Schmitt M., Hikida Y., Radev S. T., Sadlo F., & **Bürkner P. C.** (in review). The Simplex Projection: Lossless Visualization of 4D Compositional Data on a 2D Canvas. *ArXiv preprint*.
- 156) Schulze, L., Bohländer, J., **Bürkner, P. C.**, Röpke, S., Renneberg, B., & Zetsche, U. (in review). Negligible associations between cognitive control of negative information and daily emotion regulation in borderline personality disorder and major depression. *PsyArXiv preprint*.
- 155) Walter, V., Herold, J., Feuchter, S. A., Thomae, S., Venohr, M., Vogelsberg, A., Kilic, M., Berner, F., Nanz, L., Leiter-Stoeppke, U., Sinnberg, T., Schuerch, C. M., Ulmer, A., **Bürkner, P. C.**, & Flatz, L. (in review). High immune receptor clonality in melanoma-draining lymph nodes associates with immune dysfunction and poor survival. *bioRxiv preprint*.

Accepted

- 154) Bracher N., Kühmichel L., Ivanova D. R., Intes X., **Bürkner P. C.**, & Radev S. T. (accepted). JADAI: Jointly Amortizing Adaptive Design and Bayesian Inference. *Proceedings of the International Conference on Machine Learning*.

- 153) Hu N., **Bürkner P. C.**, & Arvaniti A. (accepted). A tutorial on Bayesian multivariate linear mixed-effect regression using brms. *ournal of Speech, Language, and Hearing Research*.
- 152) Schmitt, M., **Bürkner P. C.**, Köthe U., & Radev S. T. (accepted). Detecting Model Misspecification in Amortized Bayesian Inference with Neural Networks: An Extended Investigation. *International Journal of Computer Vision*.
- 2026 151) **Bürkner P. C.**, Schmitt M., & Radev S. T. (2026). Simulations in Statistical Workflows. *Philosophical Transactions A*. doi:10.1098/rsta.2024.0616
- 150) Gelman, A., Vehtari, A., McElreath, R., Simpson, D., Margossian, C. C., Yao, Y., Kennedy, L., Gabry, J., **Bürkner, P. C.**, Modrák, M., & Leos Barajas, V. (2026). Bayesian Workflow. *Chapman & Hall/CRC*.
- 149) Jedhoff S., Semenova E., Raulo A., Meyer A., & **Bürkner P. C.** (2026). From Mice to Trains: Amortized Bayesian Inference on Graph Data. *Transactions in Machine Learning Research*.
- 148) Kołczyńska M., & **Bürkner P. C.** (2026). Does political trust strengthen democracy? *International Political Science Review*. doi:10.1177/01925121251405191
- 147) Kucharský Š. & **Bürkner P. C.** (2026). Amortized Bayesian Mixture Models. *Statistics and Computing*. doi:10.1007/s11222-026-10911-y
- 146) Li C., Vehtari A., **Bürkner P. C.**, Radev S. T., Acerbi L., & Schmitt M. (2026). Amortized Bayesian Workflow. *Transactions in Machine Learning Research*.
- 145) Mishra A., Habermann D., Schmitt M., Radev S. T., & **Bürkner P. C.** (2026). Robust Amortized Bayesian Inference with Self-Consistency Losses on Unlabeled Data. *International Conference on Learning Representations*.
- 144) Mukherjee S., Claassen M., & **Bürkner P. C.** (2026). Hilbert space methods for approximating multi-output latent variable Gaussian processes. *Statistics and Computing*. doi:10.1007/s11222-026-10869-x
- 143) Reiser P., **Bürkner P. C.**, & Guthke A. (2026). Bayesian Surrogate Training on Multiple Data Sources: A Hybrid Modeling Strategy. *Statistics and Computing*. doi:10.1007/s11222-026-10906-9
- 142) Säilynoja T., Schmitt M., **Bürkner P. C.**, & Vehtari A. (2026). Posterior SBC: Simulation-Based Calibration Checking Conditional on Data. *Statistics and Computing*. doi:10.1007/s11222-026-10825-9
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